(877) 787-8375 **HELP CENTER**

MAT251_TW | Chapter 7: Sequences and Series

Started on	Saturday, July 31, 2021, 1:48 PM
State	Finished
Completed on	Saturday, July 31, 2021, 2:39 PM
Time taken	51 mins 7 secs
Grade	**140.00** out of 200.00 (**70**%)

Question **1**

Correct

5.00 points out of 5.00

Integrate $\int cot\, x\, dx.$

Select one:

- a. $\ln|sin\, x| + C$ ✓
- b. $-\ln|sec\, x| + C$
- c. $\ln|tan\, x| + C$
- d. $-csc^2 x + C$

Question **2**

Correct

5.00 points out of 5.00

Evaluate $\lim_{x\to 0^+}(e^x + x)^{1/x}.$

Select one:

- a. e^2 ✓
- b. 2
- c. 1
- d. e

Question **3**

Correct

5.00 points out of 5.00

Evaluate $\int tan^3 x \, sec \, x \, dx.$

Select one:

- a. $\frac{csc^3 x}{3} - csc \, x + C$
- b. $-\frac{sec^3 x}{3} - sec \, x + C$
- c. $\frac{csc^3 x}{3} + csc \, x + C$
- d. $\frac{sec^3 x}{3} - sec \, x + C$ ✓

Question **4**

Incorrect

0.00 points out of 5.00

If $\sum_{k=1}^{\infty} k\left(\frac{2}{3}\right)^k = 6$, then what does

$\sum_{k=1}^{\infty}(10k-3)\left(\frac{2}{3}\right)^k - \sum_{k=1}^{\infty}(5k-3)\left(\frac{2}{3}\right)^k$ equal?

Select one:

- a. 12
- b. 18
- c. 24 ✗
- d. 30

Question **5**

Correct

5.00 points out of 5.00

Evaluate $\lim_{x \to 0^+} x^{\tan x}$

Select one:

- a. 0
- b. e
- c. 1 ✓
- d. The limit doesn't exist.

Question **6**

Correct

5.00 points out of 5.00

What is the volume of the solid whose cross-sections are circles perpendicular to the base and whose diameters lie on the region bounded by *y* = *x*, the *x*-axis and *x* = 1?

Select one:

- a. 1
- b. $\frac{\pi}{4}$
- c. $\frac{\pi}{12}$ ✓
- d. $\frac{\pi}{3}$

Question **7**

Correct

5.00 points out of 5.00

What is the average value of $f(x) = x^2 + 1$ on $[1, 3]$?

Select one:

- a. $\frac{16}{3}$ ✓
- b. $\frac{14}{3}$
- c. 6
- d. 5

Question **8**

Incorrect

0.00 points out of 5.00

Find the volume of the solid of revolution obtained by rotating the region bounded by
$x = 1,\ x = 2,\ y = x^3,\ and\ y = 5x + 2$ about the *y*-axis.

Select one:

- a. $\frac{31\pi}{5}$
- b. $\frac{254\pi}{15}$
- c. $\frac{2\pi}{5}$ ✗
- d. $\frac{\pi}{15}$

Question **9**

Incorrect

0.00 points out of 5.00

Use the direct comparison test to determine whether $\sum_{n=1}^{\infty} \frac{n}{\sqrt{2n^4 - 1}}$ converges or diverges.

Select one:

- a. Converges by comparison with $\sum_{n=1}^{\infty} \frac{1}{\sqrt{2}} \frac{1}{n^2}$ ✗
- b. Converges by comparison with $\sum_{n=1}^{\infty} \frac{1}{\sqrt{2}} \frac{1}{n^3}$
- c. Diverges by comparison with $\sum_{n=1}^{\infty} \frac{1}{\sqrt{2}} \frac{1}{\sqrt{n}}$
- d. Diverges by comparison with $\sum_{n=1}^{\infty} \frac{1}{\sqrt{2}} \frac{1}{n}$

Question **10**

Correct

5.00 points out of 5.00

Evaluate $\lim_{x\to 0}(cos\, x)^{1/x^2}$.

Select one:

- a. 0
- b. $e^{1/2}$
- c. $-\frac{1}{2}$
- d. $e^{-1/2}$ ✓

Question **11**

Correct

5.00 points out of 5.00

Evaluate the integral $\int \frac{x^2}{\sqrt{1+x^3}}dx$ using $u = 1 + x^3$

Select one:

- a. $\frac{2}{3}\sqrt{1+x^3} + C$ ✓
- b. $2\sqrt{1+x^3} + C$
- c. $\frac{2}{3}x^3\sqrt{1+x^3} + C$
- d. $\frac{2}{3}\frac{x^3}{\sqrt{4x+x^4}} + C$

Question **12**

Correct

5.00 points out of 5.00

Evaluate $\lim_{x\to 0}(1+2x+x^2)^{1/x}$.

Select one:

- a. 2
- b. e^2 ✓
- c. ∞
- d. 1

Question **13**

Correct

5.00 points out of 5.00

Find an upper bound for the sequence

$$\left\{\frac{sin\ n}{n}\right\}_{n=1,2,...}.$$

Select one:

- a. 1 ✓
- b. $\frac{1}{n}$
- c. 0
- d. -1

Question **14**

Incorrect

0.00 points out of 5.00

What is the volume, as determined by the washer method, of the solid of revolution obtained by rotating the region bounded by $y = x$ and $y = x^2$ around the x-axis?

Select one:

- a. $\frac{\pi}{6}$
- b. $\frac{2\pi}{5}$
- c. $\frac{2\pi}{3}$ ✗
- d. $\frac{2\pi}{15}$

Question **15**

Correct

5.00 points out of 5.00

Find the derivative of

$$y = \ln\left[tanh\left(\frac{x}{4}\right)\right].$$

Select one:

- a. $\frac{1}{2\ cosh\ (x/2)}$
- b. $\frac{1}{2\ sinh\ (2x)}$
- c. \(\frac{1}{2\,sinh\,(x/2)} \) ✓
- d. \(\frac{1}{2\,sinh\,(x/4)} \)

Question **16**

Correct

5.00 points out of 5.00

Use the direct comparison test to

determine whether $\displaystyle\sum\limits_{n=2}^ \infty \frac{ln\,n}{n}$ converges or diverges.

Select one:

- a. Converges by comparison with $\displaystyle\sum\limits_{n=2}^ \infty \frac{1}{n\,ln\,n}$.
- b. Converges by comparison with $\displaystyle\sum\limits_{n=2}^ \infty \frac{1}{n}$.
- c. Diverges by comparison with $\displaystyle\sum\limits_{n=2}^ \infty \frac{1}{ln\,n}$.
- d. Diverges by comparison with $\displaystyle\sum\limits_{n=2}^ \infty \frac{1}{n}$. ✓

Question **17**

Correct

5.00 points out of 5.00

Which of the following statements about hyperbolic functions is **not** correct?

Select one:

- a. $csch\,x = \frac{1}{sinh\,x}\\ \quad\quad \,\,= \frac{1}{e^x-e^{-x}}$ ✓
- b. $sech\,x = \frac{1}{cosh\,x}\\ \quad\quad\,\, = \frac{2}{e^x+e^{-x}}$
- c. $At\,\, x = 0, \,\, cosh\,x=1, \,and\,\,sinh\,x = 0$
- d. $tanh\,x = \frac{e^x-e^{-x}}{e^x+e^{-x}}$

Question **18**

Incorrect

0.00 points out of 5.00

Use the direct comparison test to determine

whether $\displaystyle\sum\limits_{n=1}^ \infty \frac{n}{5n^3+2}$ converges or diverges.

Select one:

- a. $\displaystyle\sum\limits_{n=1}^ \infty \frac{n}{5n^3+2}$ converges by comparison with $\displaystyle\sum\limits_{n=1}^ \infty \frac{1}{n^2}$.
- b. $\displaystyle\sum\limits_{n=1}^ \infty \frac{n}{5n^3+2}$ converges by comparison with $\displaystyle\sum\limits_{n=1}^ \infty \frac{1}{n^3}$.
- c. $\displaystyle\sum\limits_{n=1}^ \infty \frac{n}{5n^3+2}$ diverges by comparison with $\displaystyle\sum\limits_{n=1}^ \infty \frac{1}{n^2}$. ✗
- d. $\displaystyle\sum\limits_{n=1}^ \infty \frac{n}{5n^3+2}$ diverges by comparison with $\displaystyle\sum\limits_{n=1}^ \infty \frac{1}{n^3}$.

Question **19**

Incorrect

0.00 points out of 5.00

The deflection of a hardwood beam of length L is given by

$D(x) = \frac{9}{4} x^4-7Lx^3+5L^2x^2$

where x is the distance from the fixed end of the beam. Which of the following values of x yields the maximum deflection?

Select one:

- a. 0
- b. $\frac{2L}{3}$
- c. $\frac{5L}{3}$
- d. $\frac{2L}{3}\,\,\,and\,\,\, \frac{5L}{3}$ ✗

Question **20**

Correct

5.00 points out of 5.00

For which of the following values of \(c \) will \(y = x + \frac{c}{x} \) have a relative maximum at \(x = -5 \)?

Select one:

- a. −25
- b. −5
- c. 5
- d. 25 ✓

Question **21**

Incorrect

0.00 points out of 5.00

What is the volume of the solid whose cross- sections are squares perpendicular to the x-axis and with one side on the region bounded by the curves \(y =sinx \) and the x-axis on the interval \([0, \pi/2] \)?

Select one:

- a. \(\frac{\pi}{2} \)
- b. \(\frac{\pi}{4} \)
- c. \(\frac{\pi}{5} \)
- d. \(\frac{\pi}{3} \) ✗

Question **22**

Incorrect

0.00 points out of 5.00

Which of the following identities correctly describes \(sech^2x \) in terms of \(tanh\,x \)?

Select one:

- a. \(sech^2x = 1-tanh^2x \)
- b. \(sech^2x = 1 + tanh^2x \)
- c. \(sech^2x = tanh^2 x − 1 \) ✗
- d. \(sech^2x = 1 − tanh\,x \)

Question **23**

Correct

5.00 points out of 5.00

Find the volume of the solid of revolution obtained by rotating the region bounded by \(y =x +1 \,\, and\,\, y =x^2 +1 \) about the *x*-axis.

Select one:

- a. \(\frac{7\pi}{5} \)
- b. \(\frac{\pi}{15} \)
- c. \(\frac{7\pi}{15} \) ✓
- d. \(\frac{7\pi}{30} \)

Question **24**

Correct

5.00 points out of 5.00

Evaluate \(\displaystyle \lim_{x \to \infty}(x - \sqrt{x^2+x}) \).

Select one:

- a. −1
- b. −1/2 ✓
- c. 0
- d. −∞

Question **25**

Correct

5.00 points out of 5.00

Evaluate \(\displaystyle \int \frac{x}{x^2-2x+1} dx \).

Select one:

- a. \(\ln|x-1|+\frac{2}{x-1}+C \)
- b. \(\ln|x-1|+\frac{1}{x-1}+C \)
- c. \(\ln|x-1|-\frac{2}{x-1}+C \)
- d. \(\ln|x-1|-\frac{1}{x-1}+C \) ✓

Question **26**

Correct

5.00 points out of 5.00

Which of the following is the derivative of $f(x) = e^{\cosh(x)}$?

Select one:

- a. $e^{\cosh\, x}$
- b. $e^{\cosh\,x}(\cosh\,x)$
- c. $\frac{e^{\cosh\,x}}{\sinh\,x}$
- d. $e^{\cosh\,x}(\sinh\,x)$ ✓

Question **27**

Correct

5.00 points out of 5.00

Find $\frac{d}{dx}\bigg[\cosh\,(e^x)\bigg]$.

Select one:

- a. $\cosh(e^x)$
- b. $e^x\,\sinh(e^x)$ ✓
- c. $e^x\,\cosh(e^x)$
- d. $-e^x\,\sinh(e^x)$

Question **28**

Incorrect

0.00 points out of 5.00

Evaluate the integral $\displaystyle \int_{1}^{3} \dfrac{\sqrt{4+x^2}}{x}dx$ using the integration table below.

Table of Integrals

$$\int \frac{x\,dx}{a+bx} = \frac{1}{b^2}\left(bx - a\ln|a+bx|\right) + C$$

$$\int \frac{dx}{x(a+bx)} = \frac{1}{a}\ln\left|\frac{x}{a+bx}\right| + C$$

For $a > 0$:

$$\int \frac{\sqrt{a^2+x^2}}{x}dx = \sqrt{a^2+x^2} - a\ln\left|\frac{a+\sqrt{a^2+x^2}}{x}\right| + C$$

$$\int \frac{dx}{x\sqrt{a^2+x^2}} = -\frac{1}{a}\ln\left|\frac{a+\sqrt{a^2+x^2}}{x}\right| + C$$

$$\int \frac{dx}{x^2\sqrt{a^2+x^2}} = -\frac{\sqrt{a^2+x^2}}{a^2x} + C$$

$$\int \frac{dx}{x^2\sqrt{x^2-a^2}} = \frac{\sqrt{x^2-a^2}}{a^2x} + C$$

$$\int \frac{x^2\,dx}{\sqrt{a^2-x^2}} = -\frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\arcsin\frac{x}{a} + C$$

$$\int \frac{x^2\,dx}{\sqrt{x^2-a^2}} = \frac{x}{2}\sqrt{x^2-a^2} + \frac{a^2}{2}\ln\left|x+\sqrt{x^2-a^2}\right| + C$$

Select one:

- a. $\dfrac{\sqrt{13}}{3}-\sqrt{5}$
- b. $\dfrac{1}{2}\ln(2+\sqrt{5})-\dfrac{1}{2}\ln\dfrac{2+\sqrt{13}}{3}$
- c. $\sqrt{13}-2\ln\dfrac{2+\sqrt{13}}{3}+\sqrt{5}-2\ln(2+\sqrt{5})$ ✗
- d. $\sqrt{13}-2\ln\dfrac{2+\sqrt{13}}{3}-\sqrt{5}+2\ln(2+\sqrt{5})$

Question **29**

Correct

5.00 points out of 5.00

Evaluate the integral $\displaystyle \int 6 \frac{1}{(16-x^2)^{3/2}}dx$.

Select one:

- a. $\frac{3}{8}\frac{x}{\sqrt{16-x^2}}$
- b. $\frac{3}{8}\frac{x}{\sqrt{16-x^2}} +C$ ✓
- c. $\frac{3}{4}\frac{x}{\sqrt{16-x^2}}$
- d. $\frac{3}{4}\frac{x}{\sqrt{16-x^2}} +C$

Question **30**

Incorrect

0.00 points out of 5.00

Find the derivative of

$y = \bigg(\frac{1}{2}\bigg)(\sinh\,2x-2x)$.

Select one:

- a. 0
- b. $sin^2x + cosh^2x$
- c. $2\,sinh^2x$
- d. $2\,cosh^2x$ ✗

Question **31**

Correct

5.00 points out of 5.00

Evaluate $\displaystyle\lim_{x \to \infty} \left(1+\frac{5}{x}\right)^x$.

Select one:

- a. 5
- b. 1
- c. e
- d. e^5 ✓

Question **32**

Correct

5.00 points out of 5.00

Evaluate $\displaystyle \int \cot\,x\,dx$.

Select one:

- a. $\frac{1}{\sin^2 x}+C$
- b. $-\ln|\sin\,x| +C$
- c. $\ln|\sin\,x| +C$ ✓
- d. $-\frac{1}{\sin^2x} +C$

Question **33**

Correct

5.00 points out of 5.00

What is $arccosh(1)$ equal to?

Select one:

- a. 0 ✓
- b. 1
- c. $\ln 2$
- d. e

Question **34**

Incorrect

0.00 points out of 5.00

Evaluate $\displaystyle \int_{-2}^{3}\frac{x}{(4+x^2)^{3/2}} dx$.

Select one:

- a. $\dfrac{1}{13}\sqrt{13}+\dfrac{1}{4} \sqrt{2}$
- b. $-\dfrac{1}{13}\sqrt{13}-\dfrac{1}{4}\sqrt{2}$
- c. $\dfrac{1}{13}\sqrt{13}-\dfrac{1}{4}\sqrt{2}$ ✗
- d. $-\dfrac{1}{13}\sqrt{13}+\dfrac{1}{4}\sqrt{2}$

Question **35**

Correct

5.00 points out of 5.00

When a particle is located at a distance x ft from the origin, a force of $3x^2 - 2x + 10$ pound acts on it. How much work is done in moving it from $x = 1$ to $x = 5$?

Select one:

- a. 140 ft-lb ✓
- b. 130 ft-lb
- c. 120 ft-lb
- d. 110 ft-lb

Question **36**

Correct

5.00 points out of 5.00

Evaluate $\displaystyle\lim_{x \to 1^+}\bigg(\frac{4}{x^2-1}-\frac{x}{x-1}\bigg)$.

Select one:

- a. $-\infty$
- b. 0
- c. 1
- d. ∞ ✓

Question **37**

Correct

5.00 points out of 5.00

Compute the arc length of the curve

$f(x) = \ln(1+x^2) - \frac{1}{16}x^2 - \frac{1}{8}\ln x,\ \ x \in[1, 2]$.

Select one:

- a. $\ln 9 + \frac{3}{16} + \frac{7}{8}\ln 2$
- b. $\ln 5 + \frac{3}{16} - \frac{7}{8}\ln 2$ ✓
- c. $\ln 5 + \frac{3}{16}$
- d. $\ln 9 - \frac{3}{16}$

Question **38**

Correct

5.00 points out of 5.00

What is the result of applying the integral test to the series $\displaystyle\sum\limits_{n=2}^ \infty \frac{1}{n\,ln\,n}$?

Select one:

- a. The series converges.
- b. The series diverges. ✓
- c. You cannot apply the integral test because the a_n terms are not decreasing.
- d. You cannot apply the integral test because the a_n terms are not all positive.

Question **39**

Incorrect

0.00 points out of 5.00

Find all possible values of s for which the series $\displaystyle\sum\limits_{n=1}^ \infty \frac{ln\,n}{n^s}$ converges.

Select one:

- a. $s > 1$
- b. $s > 0$ ✗
- c. $s < 0$
- d. $s \geq 2$

Question **40**

Correct

5.00 points out of 5.00

Find the derivative of $y = \dfrac{x^2+1}{x^2-1}$.

Select one:

- a. $\dfrac{-4x}{(x^2-1)^2}$ ✓
- b. 1
- c. $\dfrac{4x^3}{(x^2-1)^2}$
- d. $\dfrac{4x}{(x^2-1)^2}$

Quiz navigation

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

26 27 28 29 30 31 32 33 34 35 36 37 38 39 40

Show one page at a time

Finish review

Questions?

Please contact our Student Advisors at (877) 787-8375, Live Chat, or create a Support Request

Copyright © 2021 StraighterLine.
Privacy Policy Terms of Use